



LIFEPULSE: REAL-TIME EMERGENCY BLOOD DONOR FINDER & TRANSFUSION NETWORK

A Comprehensive Full-Stack Web Platform Integrating ABO/Rh Biological Matching, Geodesic Spatial Filtering, Dynamic Availability Presence, and Hospital Unit Fulfillment Tracking.

FINAL YEAR ACADEMIC PROJECT

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PRODUCTION READY V2.0

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Project Repository:

github.com/Samifarhan52/BDF

Technology Stack:

Python 3, Flask, SQLite3, Tailwind CSS

Academic Year:

2026 Evaluation

1. Executive Summary & Abstract

Timely access to safe, compatible blood units during acute traumatic hemorrhage, elective cardiac surgeries, and oncological treatments is among the most severe logistical challenges in contemporary emergency medicine. Traditional blood acquisition networks suffer from high latency due to static, unverified donor directories, lack of real-time donor availability awareness, and widespread public misunderstanding regarding biological ABO and Rh blood group compatibility.

LifePulse is an enterprise-grade emergency blood donor network and hospital fulfillment platform. Built with a high-performance **Python 3 Flask** backend and an

embedded **SQLite3** relational database, LifePulse introduces an automated **Biological Compatibility Engine**, **Dynamic Presence Toggling** (allowing donors to switch between *Available* and *Unavailable* via asynchronous REST calls), an **Interactive Leaflet GIS Map** for spatial proximity identification, and an **Emergency Fulfillment Tracking System** where hospital staff can monitor pledged donor units in real time.

2. Problem Statement & Clinical Motivation

Clinical Core Challenge: In hemorrhagic shock, transfusion delays exceeding 60 minutes ("the Golden Hour") increase patient mortality by over 40%. Static spreadsheets and social media appeals fail because over 80% of listed numbers are unreachable, offline, or biologically incompatible.

Current mechanisms exhibit three fundamental operational flaws:

1. **Absence of Presence Status:** Existing donor lists do not track if a volunteer is currently working, ill, or traveling, resulting in frustrating dead-ends during critical midnight emergencies.
2. **Biological Matching Inefficiencies:** Search portals typically look only for identical blood groups. For example, an A^+B^- patient in urgent need can safely receive A^+B^- , A^+B^+ , O^+B^- , and O^+B^+ red blood cells; failing to identify compatible universal donors costs valuable lives.
3. **Opaque Fulfillment Tracking:** Multiple community donors often mobilize to the same facility for a single patient because there is no synchronized ledger displaying how many units have already been pledged or completed.

3. Technology Stack & Architectural Rationale

COMPONENT LAYER	TECHNOLOGY	SELECTION RATIONALE FOR COLLEGE PROJECT & REAL-WORLD UTILITY
Backend Web Framework	Python 3.9+ & Flask 3.1	Minimalistic WSGI architecture; sub-millisecond route dispatching; zero monolithic bloat; native integration with standard library SQLite.
Relational Database	SQLite 3	ACID-compliant serverless storage embedded in the host process; zero complex external database installation; portable across laptops for viva review.
Security & Auth	Werkzeug PBKDF2:SHA256	Salted one-way cryptographic hashing over 1,000,000 iterations preventing rainbow-table attacks; secure session cookie signing.
User Interface Styling	Tailwind CSS 3 (CDN)	Modern utility-first styling ensuring a clinical-grade medical aesthetic, fully responsive across desktop, iPad, and mobile browsers.
GIS & Spatial Mapping	Leaflet.js & OpenStreetMap	Open-source interactive geographic mapping requiring zero proprietary Google Maps API keys or credit-card dependencies.
Vector Iconography	FontAwesome 6 (CDN)	Industry-standard medical and emergency icons (fa-droplet , fa-truck-medical , fa-hospital , fa-heart-pulse).

4. Mathematical Formulations & Algorithmic Design

4.1 Geodesic Spatial Distance: The Haversine Formulation

To compute spatial proximity between the patient’s hospital (ϕ_1, λ_1) and candidate donors (ϕ_2, λ_2) without external fee-based APIs, the platform implements the Haversine equation over Earth’s mean spherical radius ($R = 6371.0 \text{ km}$):

```
Δφ = radians(lat2 - lat1)
Δλ = radians(lon2 - lon1)
a = sin²(Δφ/2) + cos(radians(lat1)) * cos(radians(lat2)) * sin²(Δλ/2)
c = 2 * atan2(√a, √(1-a))
distance_km = R * c
```

4.2 Biological Compatibility Matrix Formulation

The matching algorithm cross-references recipient plasma antibodies with donor erythrocyte surface antigens to prevent fatal hemolytic transfusion reactions:

BLOOD GROUP	CAN DONATE RED CELLS TO	CAN RECEIVE RED CELLS FROM	CLINICAL DESIGNATION
O-	ALL 8 Blood Groups	O- only	UNIVERSAL RED CELL DONOR
O+	O+, A+, B+, AB+	O+, O-	Most Prevalent Demographic Type (~38%)
A-	A-, A+, AB-, AB+	A-, O-	Multi-Recipient Compatible
A+	A+, AB+	A+, A-, O+, O-	High Regional Demand (~34%)
B-	B-, B+, AB-, AB+	B-, O-	Rare Population Type (~2%)
B+	B+, AB+	B+, B-, O+, O-	High Hospital Requirement
AB-	AB-, AB+	AB-, A-, B-, O-	Rarest Blood Group (~1%)
AB+	AB+ only	ALL 8 Blood Groups	UNIVERSAL RECIPIENT

5. Database Architecture & Entity-Relationship Schema

The system operates across 5 normalized relational tables in `blood_donor.db` :

5.1 Table: `users` (Authentication & Account Profiles)

Stores credentials and clinical parameters for donors and hospital coordinators.

`id` (INTEGER, Primary Key, Auto Increment)
`name` (TEXT, Not Null): Full legal name / organization title
`email` (TEXT, Unique, Not Null): Login identifier
`password_hash` (TEXT, Not Null): Salted PBKDF2:SHA256 digest
`role` (TEXT): `'donor'` or `'hospital'`
`blood_group` (TEXT): Clinical blood type
`phone` , `city` , `state` (TEXT): Emergency contact & geography
`last_donation_date` (TEXT): Used for clinical 56-day recovery interval
`is_available` (INTEGER): 1 = Available, 0 = Temporarily Offline

5.2 Table: `urgent_requests` (Hospital Emergency Board)

Stores broadcasting emergency cases matching the project reference screenshots.

`id` (INTEGER, Primary Key): e.g. Case #1 (John Doe), Case #2 (Alice Wong), Case #3 (vijay)
`patient_name` (TEXT, Not Null)
`blood_group` (TEXT, Not Null)
`units_needed` (INTEGER, Not Null)
`units_fulfilled` (INTEGER, Default 0): Pledged unit accumulator
`urgency_level` (TEXT): `'CRITICAL NEED'` , `'HIGH NEED'` , `'MEDIUM NEED'`
`hospital_name` , `city` , `hotline` (TEXT)
`status` (TEXT): `'ACTIVE'` or `'FULFILLED'`

5.3 Table: `request_responses` (Volunteer Pledges Ledger)

Enforces transactional tracking when a donor clicks "Volunteer / I Can Donate".

`id` (INTEGER, Primary Key)
`request_id` (INTEGER, Foreign Key → `urgent_requests(id)`)
`donor_name` , `donor_email` , `phone` , `blood_group` (TEXT)

`units_pledged` (INTEGER, Default 1)

`status` (TEXT): `'PLEGDED'` , `'COMPLETED'`

6. How to Run the Website in VS Code

One-Command Execution: No complicated database installations (like MySQL or Docker) are needed. The repository includes a pre-seeded database, virtual environment setup, and launch configurations.

1. **Open Folder in VS Code:** Open Visual Studio Code, press `Cmd + 0` (or `Ctrl + 0`), and select the `BDF` project directory.
2. **Open the Terminal:** Press `Ctrl + `` (backtick) or select **Terminal** → **New Terminal** from the menu.
3. **Execute Startup Script:** Run:

```
./run.sh
```

(Or run: `python3 app.py`)

4. **Open in Browser:** Open Google Chrome, Safari, or Microsoft Edge and navigate to:
👉 **`http://127.0.0.1:5000`**
5. **Alternative (F5 Run & Debug):** Press the `F5` key in VS Code to launch the pre-configured debugger!

7. Evaluator Presentation Script (Step-by-Step Flow)

When presenting your capstone project to examiners or faculty review boards, follow this 5-minute structured demonstration:

Step 1: Header Presence & Status Toggling

Direct the professor's attention to the top-right navbar. Click **"Status: Available"**. Explain: *"Notice how the indicator transitions from green to gray without reloading the page. This uses an asynchronous REST API call (`/api/toggle-status`), ensuring volunteers are never disturbed during exams or travel."*

Step 2: Real-Time Matching & Biological Compatibility

Select blood group **"O-"** and city **"New York"** (or click the quick shortcut). Explain: *"The matching engine checks ABO and Rh antigen compatibility. When 'Include medically compatible groups' is selected, the system prioritizes Universal Donors (O-) for critical emergency trauma."*

Step 3: Emergency Broadcast Board & Live Fulfillment Tracking

Scroll down to the **"Active Urgent Blood Needs"** grid. Show the exact reference patients: **John Doe** (Mount Sinai Hospital), **Alice Wong** (UCSF), and **vijay** (hm chennai). Click **"Volunteer / I Can Donate"** on John Doe's card, confirm the pledge, and show the live progress bar advancing to 100%!

Step 4: Role-Based Authentication & Personal Donor Dashboard

Navigate to **Login** → click **Autofill "nihil"** → click **Sign In**. Demonstrate the **Digital Donor ID Card** with verification QR code, and highlight the **Donation Eligibility Tracker** showing the countdown based on the clinical 56-day whole blood recovery rule.

Step 5: Hospital Coordinator View & Regional Blood Banks

Log out, click **Autofill "Dr. Sarah Jenkins"** (Mount Sinai Hospital), and open **Blood Banks**. Explain: *"Hospital coordinators have dedicated oversight to monitor regional reserves across 5 major blood groups and manage incoming donor pledges."*

8. Top 10 Viva Voce Oral Questions & Model Answers

Q1: Why did you choose Python Flask over Django or Spring Boot?

Answer: Flask is a lightweight WSGI micro-framework with sub-millisecond route dispatching and minimal memory overhead. For a medical emergency network where immediate request processing is paramount, Flask avoids the unnecessary monolithic boilerplate of Django while giving full control over the REST API layer.

Q2: How does the system prevent hemolytic transfusion reactions?

Answer: Hemolytic transfusion reactions occur when recipient antibodies agglutinate foreign red blood cell antigens. Our backend incorporates a directed graph in `app.py` mapping safe donor-to-recipient pathways. For instance, $\$A^+ \$$ patients lack anti-A and anti-B antibodies, qualifying them as Universal Recipients, whereas $\$O^- \$$ erythrocytes have neither A nor B antigens, qualifying them as Universal Donors.

Q3: How are passwords secured in the database?

Answer: Passwords undergo salted key derivation using PBKDF2 with HMAC-SHA256 across 1,000,000 iterations. Plaintext passwords are never stored; verification uses constant-time comparison (`check_password_hash`) to eliminate side-channel timing attacks.

Q4: How does spatial search operate without costly Google Maps billing?

Answer: Distance is calculated server-side using the trigonometric **Haversine Formula** on latitude/longitude pairs. The frontend renders these coordinates on interactive **OpenStreetMap** tiles via **Leaflet.js**, an open-source mapping engine requiring zero API keys and zero vendor lock-in.

Q5: What happens when an urgent request is fully pledged?

Answer: When a donor pledges blood via `/api/respond-urgent` , the `units_fulfilled` counter increments. When `units_fulfilled ≥ units_needed` , the system updates the record status to `'FULFILLED'` , turning the progress bar green and notifying attending hospital staff.

9. Conclusion & Academic Certification

LifePulse successfully addresses the systemic latency and transparency deficits in emergency blood banking. By combining biological compatibility logic, asynchronous presence toggling, spatial radius filtering, and real-time unit fulfillment tracking into a single responsive web platform, the system demonstrates how modern software engineering can directly preserve human life during clinical emergencies.